

Car Number 19

University National polytechnic school of Algiers

Team Name Enp Racing Team

| System      |                             | Cost         | Carbon Footprint (Kg CO2e) |
|-------------|-----------------------------|--------------|----------------------------|
| BR          | Brake System                | 1,188.000000 | -                          |
| FR          | Chassis and Body            | 766.000000   | -                          |
| DR          | Drivetrain                  | -            | -                          |
| EN          | Engine and Tractive System  | 2,093.561200 | 61.417632                  |
| EL          | Grounded Low Voltage System | 247.000000   | -                          |
| MS          | Miscellaneous               | 258.000000   | -                          |
| ST          | Steering System             | 548.000000   | -                          |
| SU          | Suspension System           | 2,224.000000 | -                          |
| WT          | Wheels and Tyres            | 1,512.704000 | -                          |
| GRAND TOTAL |                             | 8,837.265200 | 61.417632                  |

| System       | Assembly    | Part                                              | M/B | P/M/F/T | Comments                                  | Qty         | Cost      | Cost | Comments | Carbon Footprint (Kg CO2e) | Carbon | Comments | ID          |
|--------------|-------------|---------------------------------------------------|-----|---------|-------------------------------------------|-------------|-----------|------|----------|----------------------------|--------|----------|-------------|
|              |             |                                                   |     |         |                                           | (per piece) |           |      |          |                            |        |          |             |
| BRAKE SYSTEM |             |                                                   |     |         | Total                                     |             | 1,188.00  |      |          | 0                          |        |          |             |
| BR           | Balance Bar |                                                   |     |         |                                           |             | 47.00     |      |          | 0.000000                   |        |          |             |
|              |             | Remote cable bias adjuster OBPCB009 (1700 mm)     | b   |         |                                           | 1           | 47.000000 |      |          |                            |        |          | BR-0010-014 |
| BR           | Brake Discs |                                                   |     |         |                                           |             | 220.00    |      |          | 0.000000                   |        |          |             |
|              |             | Rear Brake Discs                                  | b   |         |                                           | 2           | 50.000000 |      |          |                            |        |          | BR-0002-004 |
|              |             | Front Brake Discs                                 | b   |         |                                           | 2           | 60.000000 |      |          |                            |        |          | BR-0002-005 |
| BR           | Brake Fluid |                                                   |     |         |                                           |             | 0.00      |      |          | 0.000000                   |        |          |             |
|              |             | Brake Fluid                                       | b   |         |                                           | 1           |           |      |          |                            |        |          | BR-0003-006 |
| BR           | Brake Lines |                                                   |     |         |                                           |             | 337.00    |      |          | 0.000000                   |        |          |             |
|              |             | Stainless braided brake hose -3 AN (PTFE-lined)   | b   |         | 12.5 m total brake line                   | 1           | 26.000000 |      |          |                            |        |          | BR-0004-007 |
|              |             | Banjo bolt -3 AN (OBP-AN3-BB)                     | b   |         | M10×1.0 to -3 AN. Caliper inlet fittings. | 4           | 4.000000  |      |          |                            |        |          | BR-0004-008 |
|              |             | Brake hose P-clip (OBPPC-9)                       | b   |         | Hose routing clips along chassis.         | 15          | 1.000000  |      |          |                            |        |          | BR-0004-009 |
|              |             | Copper crush washer                               | b   |         | Two washers per banjo joint.              | 8           | 3.000000  |      |          |                            |        |          | BR-0004-010 |
|              |             | Brake fitting -3 AN male                          | b   |         | Hose end fittings                         | 6           | 7.000000  |      |          |                            |        |          | BR-0004-011 |
|              |             | Brake fitting -3 AN female                        | b   |         | Bulkhead/union fittings.                  | 4           | 4.000000  |      |          |                            |        |          | BR-0004-012 |
|              |             | Reservoir feed pipe, stainless braided (OBPST002) | b   |         | Master cylinder reservoir feed line.      | 2           | 20.000000 |      |          |                            |        |          | BR-0004-013 |
|              |             | Banjo Hose Ends St OBP-AN3-10BM-90                | b   |         |                                           | 4           | 17.000000 |      |          |                            |        |          | BR-0004-018 |
|              |             | Female Y Piece OBP-YA-3F                          | b   |         |                                           | 2           | 22.000000 |      |          |                            |        |          | BR-0004-019 |
|              |             | T-piece OBP-AN3-TMT                               | b   |         |                                           | 2           | 13.000000 |      |          |                            |        |          | BR-0004-020 |

|                  |                       |                                                   |   |                                                                                            |   |            |          |             |
|------------------|-----------------------|---------------------------------------------------|---|--------------------------------------------------------------------------------------------|---|------------|----------|-------------|
|                  |                       | 1/8 NPT Brake Light Inline Pressure Switch (OBP-B | b |                                                                                            | 1 | 20.000000  |          | BR-0004-021 |
| BR               | Brake Master Cylinder |                                                   |   |                                                                                            |   | 100.00     | 0.000000 |             |
|                  |                       | OBP Master Cylinder                               | b | 22.2 mm bore                                                                               | 2 | 50.000000  |          | BR-0005-001 |
| BR               | Brake Pads            |                                                   |   |                                                                                            |   | 0.00       | 0.000000 |             |
|                  |                       | Wilwood GP200 Brake Pads                          | b | Two packs of 4 pads .                                                                      | 8 |            |          | BR-0006-003 |
| BR               | Calipers              |                                                   |   |                                                                                            |   | 424.00     | 0.000000 |             |
|                  |                       | Wilwood GP200 Caliper, 2-piston                   | b | 31.75 mm pistons                                                                           | 4 | 106.000000 |          | BR-0007-002 |
| BR               | Other                 |                                                   |   |                                                                                            |   | 60.00      | 0.000000 |             |
|                  |                       | Brake fluid reservoir (OBPCB004B)                 | b | Threaded reservoir, one per master cylinder.                                               | 2 |            |          | BR-0134-015 |
|                  |                       | Reservoir bracket                                 | m | Steel, laser-cut + bent . Mounts reservoirs to firewall.                                   | 1 |            |          | BR-0001-016 |
|                  |                       | 3/8 UNF male×1/8 NPT female tee adapter           | b | Ref OBP-BPS02 . Tee for brake pressure switch + brake light switch ports, one per circuit. | 2 | 30.000000  |          | BR-0134-017 |
| CHASSIS AND BODY |                       |                                                   |   | Total                                                                                      |   | 766.00     | 0        |             |
| FR               | Aerodynamic Wing      |                                                   |   |                                                                                            |   | 0.00       | 0.000000 |             |
|                  |                       | Front Wing                                        | m |                                                                                            | 1 |            |          | FR-0048-023 |
|                  |                       | Rear Wing                                         | m |                                                                                            | 1 |            |          | FR-0048-024 |
|                  |                       | Wing Supports                                     | m |                                                                                            | 2 |            |          | FR-0048-025 |
| FR               | Bodywork              |                                                   |   |                                                                                            |   | 0.00       | 0.000000 |             |
|                  |                       | Front Nose Cone                                   | m |                                                                                            | 1 |            |          | FR-0052-020 |
|                  |                       | Side Panels                                       | m |                                                                                            | 2 |            |          | FR-0052-021 |
|                  |                       | Engine Cover                                      | m |                                                                                            | 1 |            |          | FR-0052-022 |
|                  |                       | Bodywork mould — MDF panels                       | m | 9 m² total MDF 4 mm thickness, laser-cut , used as bodywork moulds                         | 1 |            |          | FR-0052-053 |
| FR               | Floor Pan             |                                                   |   |                                                                                            |   | 0.00       | 0.000000 |             |
|                  |                       | Floor pan panel                                   | m |                                                                                            | 1 |            |          | FR-0055-009 |
| FR               | Frame Jig             |                                                   |   |                                                                                            |   | 0.00       | 0.000000 |             |

|                                            |   |                                                 |   |  |             |
|--------------------------------------------|---|-------------------------------------------------|---|--|-------------|
| Frame welding jig — steel base plate       | m | S235JR steel plate 2.7 mm base for jig fixation | 1 |  | FR-0056-049 |
| Frame welding jig — cutting gabarit plates | m | S235JR plates 3.5 mm, plasma cut                | 1 |  | FR-0056-050 |
| Frame welding jig — 90° fixing plate       | m | S235JR plate 5 mm for 90° jig fixation          | 1 |  | FR-0056-051 |
| Frame welding jig — square steel tube      | m | S235JR square tube chute for jig base member    | 1 |  | FR-0056-052 |

|    |                     |  |  |        |          |
|----|---------------------|--|--|--------|----------|
| FR | Frame / Frame Tubes |  |  | 176.00 | 0.000000 |
|----|---------------------|--|--|--------|----------|

|                         |   |                                      |    |           |             |
|-------------------------|---|--------------------------------------|----|-----------|-------------|
| Steel Tube Ø31 × 3 mm   | b | Qty represents total tube length (m) | 2  | 13.000000 | FR-0057-001 |
| Steel Tube Ø27 × 1.7 mm | b | Qty represents total tube length (m) | 30 | 5.000000  | FR-0057-034 |

|    |                   |  |  |        |          |
|----|-------------------|--|--|--------|----------|
| FR | Impact Attenuator |  |  | 290.00 | 0.000000 |
|----|-------------------|--|--|--------|----------|

|                          |   |                                                                         |   |            |             |
|--------------------------|---|-------------------------------------------------------------------------|---|------------|-------------|
| Impact Attenuator        | b | Plascore Standard FSAE honeycomb IA, purchased from USA                 | 2 | 85.000000  | FR-0058-002 |
| Anti-intrusion plate     | m |                                                                         | 1 |            | FR-0058-003 |
| Adhesive LOCTITE EA 9466 | b | structural adhesive, shear strength 24 MPa for IA-to-AIP bond (T3.17.9) | 1 | 120.000000 | FR-0058-047 |

|    |                          |  |  |      |          |
|----|--------------------------|--|--|------|----------|
| FR | Mounts Integral to Frame |  |  | 0.00 | 0.000000 |
|----|--------------------------|--|--|------|----------|

|                              |   |                                        |    |  |             |
|------------------------------|---|----------------------------------------|----|--|-------------|
| Engine Mounts                | m |                                        | 6  |  | FR-0060-027 |
| Differential Mounts          | m | 16 × laser-cut S355 steel plates 10 mm | 16 |  | FR-0059-028 |
| Seat Mounts                  | m |                                        | 1  |  | FR-0059-029 |
| Harness Mounts               | m |                                        | 1  |  | FR-0059-030 |
| Radiator Mounts              | m |                                        | 1  |  | FR-0059-035 |
| Fuel Tank Mounts             | m |                                        | 1  |  | FR-0059-036 |
| Front Wing Mounting Brackets | m |                                        | 1  |  | FR-0059-037 |
| Rear Wing Mounting Brackets  | m |                                        | 1  |  | FR-0059-038 |
| Sidepod Mounting Brackets    | m |                                        | 1  |  | FR-0059-039 |
| Floor Mounting               | m |                                        | 1  |  | FR-0059-040 |

|            |                       |                                        |   |                                                                                           |                   |  |          |             |
|------------|-----------------------|----------------------------------------|---|-------------------------------------------------------------------------------------------|-------------------|--|----------|-------------|
|            |                       | Impact Attenuator Mounting             | m |                                                                                           | 1                 |  |          | FR-0059-041 |
|            |                       | Suspension Triangle Mounts             | m | laser-cut S355 steel tabs 110 mm                                                          | 48                |  |          | FR-0059-042 |
|            |                       | Damper Mounts                          | m |                                                                                           | 16                |  |          | FR-0059-043 |
|            |                       | Steering Column Mounts                 | m |                                                                                           | 2                 |  |          | FR-0059-046 |
|            |                       | Intake mounting                        | m |                                                                                           | 1                 |  |          | FR-0059-054 |
| FR         | Pedal Box             |                                        |   |                                                                                           | 300.00            |  | 0.000000 |             |
|            |                       | Pedal Box Assembly (3 Pedal System )   | b |                                                                                           | 1 300.000000<br>0 |  |          | FR-0060-015 |
| FR         | Sidepod               |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Sidepod Left                           | m |                                                                                           | 1                 |  |          | FR-0064-013 |
|            |                       | Sidepod Right                          | m |                                                                                           | 1                 |  |          | FR-0064-026 |
| DRIVETRAIN |                       |                                        |   | Total                                                                                     | 0.00              |  | 0        |             |
| DR         | Axles                 |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Drive shafts                           | b | Bought and then remanufactured to reduce weight and lengths to achieve the custom lenghts | 2 0.000000        |  |          | DR-0142-002 |
|            |                       | Splines                                | b |                                                                                           | 1                 |  |          | DR-0142-003 |
| DR         | Chain / Belt          |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Chain                                  | b |                                                                                           | 1                 |  |          | DR-0143-008 |
| DR         | CV Joints / U Joints  |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | CV joints                              | b |                                                                                           | 1                 |  |          | DR-0144-004 |
| DR         | Differential          |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Open differential — Toyota Corolla OEM | b | OEM Corolla open differential assembly including housing and internal gears               | 1                 |  |          | DR-0145-005 |
| DR         | Differential Bearings |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Bearings                               | b | SKF 6210-2RS1/C3 differential bearings                                                    | 1 0.000000        |  |          | DR-0146-007 |
| DR         | Sprocket / Pulleys    |                                        |   |                                                                                           | 0.00              |  | 0.000000 |             |
|            |                       | Rear sprocket                          | m | 10B1 46T steel sprocket — bought blank, drilled + counterbored                            | 1                 |  |          | DR-0149-010 |

|                            |                    |                                                 |   |                                                                                                                                                                                   |      |            |                              |             |                                                                |
|----------------------------|--------------------|-------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|------------------------------|-------------|----------------------------------------------------------------|
| DR                         | Other              |                                                 |   |                                                                                                                                                                                   |      | 0.00       |                              | 0.000000    |                                                                |
|                            |                    | Pro Race Hyd- b                                 |   | OBP Pro Race hydraulic-                                                                                                                                                           | 1    |            |                              |             | DR-0167-011                                                    |
|                            |                    | to-Cable                                        |   | to-cable clutch converter                                                                                                                                                         |      |            |                              |             |                                                                |
|                            |                    | Clutch                                          |   | mechanism (OBPCCA)                                                                                                                                                                |      |            |                              |             |                                                                |
|                            |                    | Converter                                       |   |                                                                                                                                                                                   |      |            |                              |             |                                                                |
| ENGINE AND TRACTIVE SYSTEM |                    |                                                 |   | Total                                                                                                                                                                             |      | 2,093.56   |                              | 61.41763188 |                                                                |
| EN                         | Air Filter         |                                                 |   |                                                                                                                                                                                   |      | 36.00      |                              | 0.000000    |                                                                |
|                            |                    | Air Filter                                      | b |                                                                                                                                                                                   | 1    | 36.000000  |                              |             | EN-0012-004                                                    |
| EN                         | Coolant            |                                                 |   |                                                                                                                                                                                   |      | 85.57      |                              | 0.000000    |                                                                |
|                            |                    | Coolant Hoses                                   | b |                                                                                                                                                                                   | 1    | 40.000000  |                              |             | EN-0017-010                                                    |
|                            |                    | Coolant                                         | b | Distilled water                                                                                                                                                                   | 1    | 3.000000   |                              |             | EN-0017-019                                                    |
|                            |                    | water pump                                      | b |                                                                                                                                                                                   | 1    | 42.568600  |                              |             | EN-0017-030                                                    |
| EN                         | Engine             |                                                 |   |                                                                                                                                                                                   |      | 939.75     |                              | 0.000000    |                                                                |
|                            |                    | Yamaha FZ6 S1 600cc Engine                      | b | Used Yamaha FZ6 S1 engine assembly, including engine block, gearbox, clutch, starter and alternator. OEM auxiliary components are listed separately when installed on the vehicle | 1    | 879.750000 |                              |             | EN-0023-001                                                    |
|                            |                    | Rubber isolators for engine mounts (caoutchouc) | b |                                                                                                                                                                                   | 6    | 10.000000  |                              |             | EN-0023-028                                                    |
| EN                         | Exhaust Manifold   |                                                 |   |                                                                                                                                                                                   |      | 34.97      |                              | 12.710333   |                                                                |
|                            |                    | Exhaust Manifold                                | m |                                                                                                                                                                                   | 1    |            |                              |             | EN-0025-013                                                    |
|                            |                    |                                                 | P | Weld                                                                                                                                                                              | 1.00 | 4.34       |                              | 0.278333    | EN-0025-013-0016                                               |
|                            |                    |                                                 | M | Steel                                                                                                                                                                             | 2.00 | 12.00      | Qty in (Kg) , Cost (€ / Kg ) | 6.150000    | Carbon footprint in ( Kg CO2e / Kg material ) EN-0025-013-0017 |
|                            |                    |                                                 | F | Bolt                                                                                                                                                                              | 8.00 | 0.60       |                              | 0.000000    | EN-0025-013-0020                                               |
|                            |                    |                                                 | P | Tube Process                                                                                                                                                                      | 1.00 |            |                              | 0.000000    | EN-0025-013-0028                                               |
|                            |                    |                                                 | P | Cutting (Manual)                                                                                                                                                                  | 0.20 | 9.14       | Qty ( hours ) , cost ( € /h) | 0.660000    | (kgCOe/h) EN-0025-013-0029                                     |
| EN                         | Fuel Filter        |                                                 |   |                                                                                                                                                                                   |      | 20.00      |                              | 0.000000    |                                                                |
|                            |                    | Fuel Filter                                     | b |                                                                                                                                                                                   | 1    | 20.000000  |                              |             | EN-0026-007                                                    |
| EN                         | Fuel Lines / Rails |                                                 |   |                                                                                                                                                                                   |      | 80.00      |                              | 0.000000    |                                                                |
|                            |                    | Fuel hoses                                      | b |                                                                                                                                                                                   | 1    | 80.000000  |                              |             | EN-0028-026                                                    |

|    |                                |   |                     |                                                                                                   |      |            |                                                             |           |                                             |  |                  |
|----|--------------------------------|---|---------------------|---------------------------------------------------------------------------------------------------|------|------------|-------------------------------------------------------------|-----------|---------------------------------------------|--|------------------|
|    | Pneumatic elbow fitting        | b |                     | Elbow fitting installed on the filler neck for filling/vent visualization line                    | 2    |            |                                                             |           |                                             |  | EN-0028-032      |
|    | Flexible hose                  | b |                     | Flexible hose connected to the filler neck to visualize filling/venting                           | 1    |            |                                                             |           |                                             |  | EN-0028-034      |
| EN | Fuel Pump                      |   |                     |                                                                                                   |      | 113.45     |                                                             |           | 0.000000                                    |  |                  |
|    | Fuel Pump                      | b |                     | OEM yamaha fz6 pump                                                                               | 1    | 113.450400 |                                                             |           |                                             |  | EN-0031-006      |
|    | Fuel resistant O-ring          | b |                     | O-ring used to seal the fuel pump interface                                                       | 1    | 0.000000   |                                                             |           |                                             |  | EN-0031-033      |
| EN | Fuel Tank - not the HV Battery |   |                     |                                                                                                   |      | 71.39      |                                                             |           | 36.415550                                   |  |                  |
|    | Fuel Tank                      | m |                     | Aluminium fuel tank with internal baffles, bottom-mounted fuel pump interface and vent valve port | 1    |            |                                                             |           |                                             |  | EN-0032-002      |
|    |                                | M | Aluminium           | Al-2017A 3 mm                                                                                     | 4.06 | 12.00      | Quantity per Kg of raw material , cost (€/Kg raw material ) | 8.240000  | carbon footprint (Kg CO2 /Kg raw material ) |  | EN-0032-002-0011 |
|    |                                | P | Weld                | TIG welding of tank panels, internal baffles and mounting interfaces                              | 0.45 | 15.58      | Qty per Hours , Cost (€/Hours)                              | 2.200000  | (Kg CO2 / hours)                            |  | EN-0032-002-0013 |
|    |                                | P | Sheet metal Process | Bending of aluminium fuel tank panels according to the drawing                                    | 0.15 | 8.18       | Qty per Hours , Cost (€/Hours)                              | 0.825000  | (Kg CO2 / hours)                            |  | EN-0032-002-0014 |
|    |                                | F | Bolt                | Threaded drain plug used to close the fuel tank drain port                                        | 4.00 | 0.70       |                                                             | 0.000000  | Bought parts                                |  | EN-0032-002-0025 |
|    |                                | F | Nut                 | Welded threaded drain bung used for fuel tank draining                                            | 4.00 | 0.25       |                                                             | 0.000000  | Bought parts                                |  | EN-0032-002-0026 |
|    |                                | F | Washer              | Sealing washer used for drain plug sealing                                                        | 1.00 | 0.10       |                                                             | 0.000000  | Bought parts                                |  | EN-0032-002-0031 |
|    |                                | F | Nut                 | Welded threaded bung used to mount the Newton VRJ6-6 fuel tank vent valve                         | 1.00 | 0.25       |                                                             | 0.000000  | Bought                                      |  | EN-0032-002-0032 |
|    |                                | P | Drilled hole        | Drilling of the hole on the filler neck for the pneumatic elbow fitting                           | 0.08 | 7.83       | €/h                                                         | 0.410000  | (kgCOe/h)                                   |  | EN-0032-002-0035 |
|    |                                | F | Bolt                | Screws used to fix the fuel pump to the fuel tank mounting ring                                   | 8.00 | 0.70       |                                                             | 0.000000  |                                             |  | EN-0032-002-0036 |
|    |                                | P | Laser Cut           | Laser cutting of tank panels, baffles, pump opening, drain hole and vent valve port               | 0.09 | 28.68      | Qty per hours ,C                                            | 11.000000 | (kgCOe/h)                                   |  | EN-0032-002-0037 |
|    | 90° filler neck tube           | m |                     | 90° tube used as the fuel tank filler neck                                                        | 1    |            |                                                             |           |                                             |  | EN-0032-031      |

|    |                                        |   |   |                                       |                                                                                                      |      |            |                                                                                                 |          |             |                  |
|----|----------------------------------------|---|---|---------------------------------------|------------------------------------------------------------------------------------------------------|------|------------|-------------------------------------------------------------------------------------------------|----------|-------------|------------------|
|    |                                        |   | M | Aluminium                             |                                                                                                      | 0.10 | 12.00      | €/kg                                                                                            | 8.240000 | kgCOe/kg    | EN-0032-031-0033 |
|    |                                        |   | P | Cutting (Manual)                      | Cutting of the filler neck tube to required length                                                   | 0.03 | 9.14       |                                                                                                 | 0.020000 |             | EN-0032-031-0034 |
| EN | Fuel Vent / Check Valve                |   |   |                                       |                                                                                                      |      | 140.00     |                                                                                                 | 0.000000 |             |                  |
|    | Newton VRJ6-6 FIA fuel tank vent valve | b |   |                                       | Newton VRJ6-6 (9/16" UNF) anti-roll + anti-surge + by-pass 200 mbar vent valve, mounted on fuel tank | 1    | 140.000000 |                                                                                                 |          |             | EN-0033-027      |
| EN | Intake Manifold                        |   |   |                                       |                                                                                                      |      | 82.54      |                                                                                                 | 9.729337 |             |                  |
|    | Plenum                                 | m |   |                                       |                                                                                                      | 1    |            |                                                                                                 |          |             | EN-0035-003      |
|    |                                        |   | M | ABS (Acrylonitrile Butadiene Styrene) |                                                                                                      | 1.00 | 14.36      |                                                                                                 | 1.780869 |             | EN-0035-003-0009 |
|    |                                        |   | P | 3D-printed                            |                                                                                                      | 1.00 | 42.10      |                                                                                                 | 4.875368 |             | EN-0035-003-0010 |
|    | Runners                                | m |   |                                       |                                                                                                      | 1    |            |                                                                                                 |          |             | EN-0035-029      |
|    |                                        |   | M | ABS                                   |                                                                                                      | 0.26 | 25.00      |                                                                                                 | 3.100000 |             | EN-0035-029-0018 |
|    |                                        |   | P | 3D printing                           |                                                                                                      | 6.87 | 2.85       | Quantity represents hours of all the process , the cost represents hourly machine rate (Euro/h) | 0.330000 | (kg CO2e/h) | EN-0035-029-0019 |
| EN | Muffler                                |   |   |                                       |                                                                                                      |      | 200.00     |                                                                                                 | 0.000000 |             |                  |
|    | Exhaust silencer                       | b |   |                                       |                                                                                                      | 1    | 200.000000 |                                                                                                 |          |             | EN-0036-011      |
| EN | Overflow Bottles                       |   |   |                                       |                                                                                                      |      | 27.57      |                                                                                                 | 0.000000 |             |                  |
|    | Coolant Overflow Bottle                | b |   |                                       |                                                                                                      | 1    | 12.770600  |                                                                                                 |          |             | EN-0039-020      |
|    | Oil Catch Tank                         | b |   |                                       | OEM fz6 engines Oil Catch tank                                                                       | 1    | 14.797900  |                                                                                                 |          |             | EN-0039-021      |
| EN | Radiator                               |   |   |                                       |                                                                                                      |      | 162.24     |                                                                                                 | 0.000000 |             |                  |
|    | Motorcycle Radiators for KTM 450       | b |   |                                       |                                                                                                      | 2    | 81.120000  |                                                                                                 |          |             | EN-0040-008      |
| EN | Radiator Fans                          |   |   |                                       |                                                                                                      |      | 27.67      |                                                                                                 | 0.000000 |             |                  |
|    | Radiator Fan                           | b |   |                                       |                                                                                                      | 1    | 27.667100  |                                                                                                 |          |             | EN-0041-009      |
| EN | Restrictor                             |   |   |                                       |                                                                                                      |      | 11.99      |                                                                                                 | 1.411010 |             |                  |
|    | Restrictor                             | m |   |                                       | 3D printed body with bought throttle body and air filter                                             | 1    |            |                                                                                                 |          |             | EN-0042-005      |
|    |                                        |   | P | 3D printing                           |                                                                                                      | 1.00 | 9.24       |                                                                                                 | 1.069932 |             | EN-0042-005-0004 |

|                             |                                                |   |   |                                                             |       |      |            |                          |          |  |                  |
|-----------------------------|------------------------------------------------|---|---|-------------------------------------------------------------|-------|------|------------|--------------------------|----------|--|------------------|
|                             |                                                |   | M | ABS                                                         |       | 0.11 | 25.00      |                          | 3.100708 |  | EN-0042-005-0008 |
| EN                          | Shields                                        |   |   |                                                             |       |      | 3.43       |                          | 1.151402 |  |                  |
|                             | Chain Guard                                    | m |   |                                                             |       | 1    |            |                          |          |  | EN-0043-023      |
|                             |                                                |   | M | S235JR steel sheet 1.5 mm                                   |       | 1.00 | 0.70       |                          | 1.023701 |  | EN-0043-023-0005 |
|                             |                                                |   | P | Laser Cut                                                   |       | 1.00 | 2.05       |                          | 0.058979 |  | EN-0043-023-0006 |
|                             |                                                |   | P | Bending                                                     |       | 1.00 | 0.68       |                          | 0.068722 |  | EN-0043-023-0007 |
| EN                          | Throttle Body                                  |   |   |                                                             |       |      | 22.00      |                          | 0.000000 |  |                  |
|                             | Throttle Body                                  | b |   | d45.5/sf45.5mm                                              |       | 1    | 22.000000  |                          |          |  | EN-0012-016      |
| EN                          | Engine Oil                                     |   |   |                                                             |       |      | 35.00      |                          | 0.000000 |  |                  |
|                             | Engine Oil                                     | b |   |                                                             |       | 1    | 35.000000  | Cost for (4L) Engine oil |          |  | EN-0136-017      |
| GROUNDED LOW VOLTAGE SYSTEM |                                                |   |   |                                                             | Total |      | 247.00     |                          | 0        |  |                  |
| EL                          | Brake Pressure Sensors                         |   |   |                                                             |       |      | 39.00      |                          | 0.000000 |  |                  |
|                             | Brake system pressure sensor X05PSA02000 P18K  | b |   | Brake system pressure sensor for shutdown circuit           |       | 1    |            |                          |          |  | EL-0071-013      |
|                             | BRAKE OVER-TRAVEL SWITCH                       | b |   |                                                             |       | 1    | 39.000000  |                          |          |  | EL-0071-015      |
| EL                          | Dash Panel                                     |   |   |                                                             |       |      | 104.00     |                          | 0.000000 |  |                  |
|                             | Stoneitech STWA050WN-11 steering wheel display | b |   | Programmable textile display integrated into steering wheel |       | 1    | 104.000000 |                          |          |  | EL-0073-002      |
| EL                          | ECM / Engine Electronics                       |   |   |                                                             |       |      | 0.00       |                          | 0.000000 |  |                  |
|                             | Exhaust Oxygen Sensor                          | b |   |                                                             |       | 1    |            |                          |          |  | EL-0074-003      |
|                             | Throttle Position Sensor                       | b |   |                                                             |       | 1    |            |                          |          |  | EL-0074-004      |
|                             | Intake Air Temperature Sensor                  | b |   |                                                             |       | 1    |            |                          |          |  | EL-0074-005      |
|                             | MAP Sensor                                     | b |   |                                                             |       | 1    |            |                          |          |  | EL-0074-006      |
|                             | Crankshaft Position Sensor                     | b |   | 5VX-81670-00                                                |       | 1    |            |                          |          |  | EL-0074-007      |
|                             | Speed Sensor                                   | b |   |                                                             |       | 1    |            |                          |          |  | EL-0074-008      |

|               |                                                  |   |                     |                                                                                    |      |            |          |                  |
|---------------|--------------------------------------------------|---|---------------------|------------------------------------------------------------------------------------|------|------------|----------|------------------|
|               | ECU Yamaha FZ6 OEM                               | b |                     |                                                                                    | 1    |            |          | EL-0074-012      |
| EL            | LV Battery                                       |   |                     |                                                                                    |      | 0.00       | 0.000000 |                  |
|               | LV Battery                                       | b |                     |                                                                                    | 1    |            |          | EL-0079-016      |
| EL            | Power Distribution System                        |   |                     |                                                                                    |      | 104.00     | 0.000000 |                  |
|               | Rectifier/Regulator Unit                         | b |                     |                                                                                    | 1    | 104.000000 |          | EL-0081-011      |
|               | Shutdown buttons M22-PVT45P / Schneider XB4BT845 | b |                     |                                                                                    | 3    |            |          | EL-0081-014      |
| EL            | Dash Gauges                                      |   |                     |                                                                                    |      | 0.00       | 0.000000 |                  |
|               | FZ6 OEM cluster                                  | b |                     | OEM Yamaha FZ6 instrument cluster (Speed / RPM / Fuel / Temp / Pressure / Battery) | 1    |            |          | EL-0154-001      |
| EL            | Other                                            |   |                     |                                                                                    |      | 0.00       | 0.000000 |                  |
|               | Radiator Fluid Temperature Sensor                | b |                     |                                                                                    | 1    |            |          | EL-0156-009      |
|               | Oil monitoring sensor                            | b |                     |                                                                                    | 1    |            |          | EL-0156-010      |
| MISCELLANEOUS |                                                  |   | Total               |                                                                                    |      | 258.00     | 0        |                  |
| MS            | Driver's Harness                                 |   |                     |                                                                                    |      | 218.00     | 0.000000 |                  |
|               | Hybrid II Formula Student Harness                | b |                     | Hybrid II black Formula Student specification harness, SFI 16.1 approved.          | 1    | 218.000000 |          | MS-0091-001      |
| MS            | Firewall                                         |   |                     |                                                                                    |      | 0.00       | 0.000000 |                  |
|               | Firewall                                         | m |                     |                                                                                    | 1    |            |          | MS-0092-007      |
|               |                                                  | M | Aluminium           | Aluminium sheet used for the firewall panel                                        | 1.00 |            | 0.000000 | MS-0092-007-0002 |
|               |                                                  | P | Laser Cut           | Laser cutting of the aluminium firewall panel profile and openings                 | 1.00 |            | 0.000000 | MS-0092-007-0003 |
|               |                                                  | P | Sheet metal Process | Bending of the aluminium firewall panel according to the required shape            | 1.00 |            | 0.000000 | MS-0092-007-0004 |
|               |                                                  | P | Drilled hole        | Drilling of holes for cable glands and firewall mounting points                    | 1.00 |            | 0.000000 | MS-0092-007-0005 |
|               |                                                  | P | Tape                | 3M Aluminum Foil Tape 425 used to seal firewall panel joints and gaps              | 1.00 |            | 0.000000 | MS-0092-007-0006 |
| MS            | Headrest / Restraints                            |   |                     |                                                                                    |      | 40.00      | 0.000000 |                  |

|                   |                              |                                |   |           |  |                                                                                                  |      |            |  |          |                  |
|-------------------|------------------------------|--------------------------------|---|-----------|--|--------------------------------------------------------------------------------------------------|------|------------|--|----------|------------------|
|                   |                              | SFI Approved High Density Foam | b |           |  | SFI approved high density foam sheet, 9 × 16 in, 2 in thick, used for the driver head restraint. | 1    | 40.000000  |  |          | MS-0094-004      |
| MS                | Seats                        |                                |   |           |  |                                                                                                  |      | 0.00       |  | 0.000000 |                  |
|                   |                              | Seat shell                     | m |           |  |                                                                                                  | 1    |            |  |          | MS-0100-003      |
|                   |                              | Seat foam                      | m |           |  |                                                                                                  | 1    |            |  |          | MS-0100-005      |
| STEERING SYSTEM   |                              |                                |   |           |  | Total                                                                                            |      | 548.00     |  | 0        |                  |
| ST                | Steering Rack                |                                |   |           |  |                                                                                                  |      | 450.00     |  | 0.000000 |                  |
|                   |                              | kaz technologie steering rack  | b |           |  | used from fs team in poland                                                                      | 1    | 450.000000 |  |          | ST-0101-012      |
|                   |                              | Rack support                   | m |           |  |                                                                                                  | 1    |            |  |          | ST-0101-013      |
| ST                | Steering Shaft               |                                |   |           |  |                                                                                                  |      | 54.00      |  | 0.000000 |                  |
|                   |                              | Steering Column                | m |           |  |                                                                                                  | 1    |            |  |          | ST-0102-002      |
|                   |                              | Double Universal Joint         | b |           |  |                                                                                                  | 1    | 50.000000  |  |          | ST-0102-003      |
|                   |                              | Cast Metal Housing             | b |           |  |                                                                                                  | 1    | 4.000000   |  |          | ST-0102-004      |
| ST                | Steering Wheel               |                                |   |           |  |                                                                                                  |      | 0.00       |  | 0.000000 |                  |
|                   |                              | Steering Wheel                 | m |           |  |                                                                                                  | 1    |            |  |          | ST-0103-005      |
| ST                | Steering Wheel Quick Release |                                |   |           |  |                                                                                                  |      | 20.00      |  | 0.000000 |                  |
|                   |                              | Quick Release                  | b |           |  |                                                                                                  | 1    | 20.000000  |  |          | ST-0104-006      |
| ST                | Tie Rods                     |                                |   |           |  |                                                                                                  |      | 24.00      |  | 0.000000 |                  |
|                   |                              | Tie Rod D16                    | m |           |  |                                                                                                  | 2    |            |  |          | ST-0105-007      |
|                   |                              | Rod Ends                       | b |           |  |                                                                                                  | 2    | 12.000000  |  |          | ST-0105-008      |
| SUSPENSION SYSTEM |                              |                                |   |           |  | Total                                                                                            |      | 2,224.00   |  | 0        |                  |
| SU                | Front A / Arms or Equivalent |                                |   |           |  |                                                                                                  |      | 272.00     |  | 0.000000 |                  |
|                   |                              | Upper Front Arm                | m |           |  |                                                                                                  | 2    |            |  |          | SU-0108-001      |
|                   |                              | Lower Front Arm                | m |           |  |                                                                                                  | 2    |            |  |          | SU-0108-002      |
|                   |                              | Rod Ends                       | b |           |  |                                                                                                  | 16   | 17.000000  |  |          | SU-0108-013      |
| SU                | Front Uprights               |                                |   |           |  |                                                                                                  |      | 0.00       |  | 0.000000 |                  |
|                   |                              | front upright                  | m |           |  | Using 4-axis CNC Milling machine                                                                 | 2    |            |  |          | SU-0109-003      |
|                   |                              |                                | M | Aluminium |  |                                                                                                  | 1.00 |            |  | 0.000000 | SU-0109-003-0010 |

|    |                             |   |               |                                                                |        |           |                  |
|----|-----------------------------|---|---------------|----------------------------------------------------------------|--------|-----------|------------------|
|    |                             | P | Machining     | CNC machining                                                  | 2.00   | 0.000000  | SU-0109-003-0011 |
|    |                             | P | Drilled hole  | Drilling of suspension and brake caliper mounting holes        | 8.00   | 0.000000  | SU-0109-003-0012 |
|    |                             | P | Reamed Hole   | Precision finishing of the wheel bearing seats / central bores | 2.00   | 0.000000  | SU-0109-003-0013 |
|    |                             | P | Tapping Holes | Tapping of brake caliper mounting holes                        | 2.00   | 0.000000  | SU-0109-003-0014 |
|    |                             | P | Hand Finish   | Chamfering, deburring and final manual finishing               | 2.00   | 0.000000  | SU-0109-003-0015 |
|    |                             | F | Bolt          |                                                                | 8.00   | 0.000000  | SU-0109-003-0016 |
|    |                             | F | Nut           |                                                                | 8.00   | 0.000000  | SU-0109-003-0017 |
|    |                             | F | Washer        |                                                                | 16.00  | 0.000000  | SU-0109-003-0018 |
| SU | Pushrods / Pullrods         |   |               |                                                                | 136.00 | 0.000000  |                  |
|    | Rod Ends                    | b |               |                                                                | 8      | 17.000000 | SU-0111-010      |
|    | spacers                     | b |               |                                                                | 1      |           | SU-0111-012      |
| SU | Rear A / Arms or Equivalent |   |               |                                                                | 0.00   | 0.000000  |                  |
|    | Upper Rear Arm              | m |               | (s235, 5 mm)                                                   | 2      |           | SU-0112-007      |
|    | Lower Rear Arm              | m |               |                                                                | 2      |           | SU-0112-008      |
|    | Rod Ends                    | b |               |                                                                | 16     |           | SU-0112-014      |
| SU | Rockers                     |   |               |                                                                | 0.00   | 0.000000  |                  |
|    | Rocker Arm                  | m |               | s235jr 6 mm                                                    | 8      |           | SU-0113-009      |
| SU | Rear Uprights               |   |               |                                                                | 0.00   | 0.000000  |                  |
|    | rear uprights               | m |               | Using 4-axis CNC Milling machine                               | 2      |           | SU-0114-004      |
|    |                             | P | Machining     | 4-axis CNC milling of rear upright body and mounting features  | 1.00   | 0.000000  | SU-0114-004-0002 |
|    |                             | P | Drilled hole  | Drilling of suspension, caliper and mounting holes             | 1.00   | 0.000000  | SU-0114-004-0003 |
|    |                             | P | Reamed Hole   | Precision finishing of bearing seat / central bore             | 1.00   | 0.000000  | SU-0114-004-0004 |
|    |                             | P | Tapping Holes | Tapping of threaded mounting holes, if required                | 1.00   | 0.000000  | SU-0114-004-0005 |
|    |                             | P | Hand Finish   | Chamfering, deburring and final manual finishing               | 1.00   | 0.000000  | SU-0114-004-0006 |
|    |                             | F | Bolt          | Bolts for suspension arms and brake caliper mount              | 4.00   | 0.000000  | SU-0114-004-0007 |

|                         |                       |   |                |                                                                       |                 |          |                  |
|-------------------------|-----------------------|---|----------------|-----------------------------------------------------------------------|-----------------|----------|------------------|
|                         |                       | F | Nut            | Nuts used with rear upright mounting bolts                            | 8.00            | 0.000000 | SU-0114-004-0008 |
|                         |                       | F | Washer         | Washers used with mounting bolts                                      | 16.00           | 0.000000 | SU-0114-004-0009 |
| SU                      | Rear Damper Assembly  |   |                |                                                                       | 996.00          | 0.000000 |                  |
|                         | Suspension Dampers    | b |                | TTx 25 MK2                                                            | 2 360.000000    |          | SU-0163-006      |
|                         | springs               | b |                | 18077-17 Ohlins bicycle shock absorber (112 N/MM)                     | 2 138.000000    |          | SU-0163-016      |
| SU                      | Front Damper Assembly |   |                |                                                                       | 820.00          | 0.000000 |                  |
|                         | Suspension Dampers    | b |                | TTx 25 MK2                                                            | 2 360.000000    |          | SU-0169-005      |
|                         | springs               | b |                | Hyperco Shock Spring (70 N/MM)                                        | 2 50.000000     |          | SU-0169-015      |
| <b>WHEELS AND TYRES</b> |                       |   |                | <b>Total</b>                                                          | <b>1,512.70</b> | <b>0</b> |                  |
| WT                      | Front Hub Assemblies  |   |                |                                                                       | 80.00           | 0.000000 |                  |
|                         | lock nut sfk KMFE 9   | b |                |                                                                       | 2 40.000000     |          | WT-0120-010      |
|                         | Circlips              | b |                | to guarantee the locking of the bearing                               | 16              |          | WT-0120-011      |
| WT                      | Front Hubs            |   |                |                                                                       | 0.00            | 0.000000 |                  |
|                         | Front Wheel Hubs      | m |                |                                                                       | 2               |          | WT-0120-006      |
|                         |                       | M | Aluminium      | 2017A round bar                                                       | 1.00            | 0.000000 | WT-0120-006-0002 |
|                         |                       | P | Machining      |                                                                       | 1.00            | 0.000000 | WT-0120-006-0003 |
|                         |                       | P | Drilled hole   | Drilling of wheel mounting and flange holes                           | 8.00            | 0.000000 | WT-0120-006-0004 |
|                         |                       | P | Reamed Hole    | Precision finishing of bearing seat / central bore                    | 1.00            | 0.000000 | WT-0120-006-0005 |
|                         |                       | P | Threading      | Threading of M45x1.5 and M65x1.5 features                             | 2.00            | 0.000000 | WT-0120-006-0006 |
|                         |                       | P | Hand Finish    | Chamfering, deburring and final manual finishing                      | 1.00            | 0.000000 | WT-0120-006-0007 |
|                         |                       | F | Nut            | SKF KMFE 9 lock nut used to retain the front hub bearing.             | 2.00            | 0.000000 | WT-0120-006-0008 |
|                         |                       | F | Bolt           | Bolts used to mount the brake disc to the front wheel hub             | 1.00            | 0.000000 | WT-0120-006-0009 |
|                         |                       | F | Nut            | Central wheel nut used to retain the wheel on the front hub.          | 1.00            | 0.000000 | WT-0120-006-0010 |
|                         |                       | F | Retaining Ring | Retaining rings used for axial retention of the bearing/hub assembly. | 1.00            | 0.000000 | WT-0120-006-0011 |

|    |                                            |                |                                                                  |          |            |          |                  |
|----|--------------------------------------------|----------------|------------------------------------------------------------------|----------|------------|----------|------------------|
|    |                                            | F Bolt         | Anti-vibration bolts used to secure the front hub/wheel assembly | 1.00     |            | 0.000000 | WT-0120-006-0012 |
| WT | Rear Hub Assemblies                        |                |                                                                  | 212.00   |            | 0.000000 |                  |
|    | Rear Wheel Hubs                            | m              | CNC machined aluminium part using lathe machine                  | 2        |            |          | WT-0123-007      |
|    |                                            | M Aluminium    | 2017A round bar                                                  | 1.00     |            | 0.000000 | WT-0123-007-0013 |
|    |                                            | P Machining    | CNC turning                                                      | 1.00     |            | 0.000000 | WT-0123-007-0014 |
|    |                                            | P Drilled hole |                                                                  | 8.00     |            | 0.000000 | WT-0123-007-0015 |
|    |                                            | P Reamed Hole  | Precision finishing of bearing seats and central bore            | 1.00     |            | 0.000000 | WT-0123-007-0016 |
|    |                                            | P Broach       | Internal spline broaching for drivetrain/shaft connection        | 1.00     |            | 0.000000 | WT-0123-007-0017 |
|    |                                            | P Threading    | External/internal threading for retaining nut, if applicable     | 1.00     |            | 0.000000 | WT-0123-007-0018 |
|    |                                            | P Hand Finish  | Chamfering, deburring and final manual finishing                 | 1.00     |            | 0.000000 | WT-0123-007-0019 |
|    |                                            | F Nut          | SKF KMFE 9 lock nut used to retain the front hub bearing.        | 1.00     |            | 0.000000 | WT-0123-007-0020 |
|    | lock nut sflk KMFE 9                       | b              |                                                                  | 2        | 50.000000  |          | WT-0123-012      |
|    | Circlips                                   | b              |                                                                  | 16       | 7.000000   |          | WT-0123-013      |
| WT | Tyres                                      |                |                                                                  | 1,100.00 |            | 0.000000 |                  |
|    | Hoosier Tyres 20.5X7.0-13 R20              | b              |                                                                  | 4        | 125.000000 |          | WT-0125-002      |
|    | Hoosier 21.0x6.5-13 W3 (or 20.0x7.5-13 W3) | b              |                                                                  | 4        | 150.000000 |          | WT-0125-009      |
| WT | Wheel Bearings                             |                |                                                                  | 120.70   |            | 0.000000 |                  |
|    | Front Wheel Bearings                       | b              |                                                                  | 4        | 16.200000  |          | WT-0128-004      |
|    | Rear Wheel Bearings                        | b              |                                                                  | 4        | 13.976000  |          | WT-0128-005      |
| WT | Wheels                                     |                |                                                                  | 0.00     |            | 0.000000 |                  |
|    | Oz Racing 13-7" 4h Mg Wheels               | b              |                                                                  | 4        |            |          | WT-0131-003      |